

Question	Answer	Marks
6(a)(i)	conversion of a.c. to d.c.	B1
6(a)(ii)	full-wave (rectification)	B1
6(a)(iii)	three diodes with correct symbols connected into the circuit	B1
	lower-left diode shown pointing towards the right and upper-left and lower-right diodes both shown pointing towards the left	B1
6(b)(i)	$\omega = 18 \text{ (rad s}^{-1}\text{)}$	C1
	$T = 2\pi / 18 = 0.35 \text{ s}$	A1

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6(b)(ii)	$V_{\text{r.m.s.}} = 34 / \sqrt{2}$ $= 24 \text{ V}$	A1
6(b)(iii)	sinusoidal 'humps' with minimum $V_{\text{OUT}} = 0$ and non-zero V_{OUT} all same sign	B1
	peak V_{OUT} shown as 34 V or –34 V	B1
	two 'humps' shown, with V_{OUT} always with same sign and $V_{\text{OUT}} = 0$ at $t = 0$, $t = 0.175 \text{ s}$ and $t = 0.350 \text{ s}$ and non-zero in between	B1
6(c)(i)	correct circuit symbol for capacitor, connected in parallel with resistor	B1
6(c)(ii)	time constant = RC	C1
	$= 56 \times 10^3 \times 12 \times 10^{-6}$ $= 0.67 \text{ s}$	A1
6(c)(iii)	$V_{\text{min}} = 34 \exp(-0.14 / 0.67)$	C1
	$= 28 \text{ V}$	A1

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